

## Project Initialization and Planning Phase

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 01 October 2025                                             |
| Team ID       | SWUID20250209075                                            |
| Project Title | Global Malnutrition Trends: A Power BI Analysis (1983–2019) |
| Maximum Marks | 3 Marks                                                     |

### Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

| Project Overview |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective        | <p>The primary objective of this project is to analyze global malnutrition trends among children under five years of age between 1983 and 2019, using Kaggle datasets. The project aims to uncover correlations between different forms of malnutrition (stunting, wasting, underweight, and overweight) and a country's economic classification (low, lower-middle, upper-middle, and high income).</p> <p>Through Power BI dashboards, the solution will present actionable insights to policymakers, researchers, and organizations, enabling them to design targeted interventions for reducing malnutrition.</p>                                                                 |
| Scope            | <ul style="list-style-type: none"> <li>• Included: <ul style="list-style-type: none"> <li>• Analysis of malnutrition indicators (stunting, wasting, underweight, overweight).</li> <li>• Comparative study across income levels (LDC, LIFD, LLDC, SIDS).</li> <li>• Historical trends (1983–2019).</li> <li>• Interactive Power BI dashboard with filters, slicers, and visuals.</li> <li>• Documentation and demo video.</li> </ul> </li> <li>• Excluded: <ul style="list-style-type: none"> <li>• Real-time data analysis.</li> <li>• Predictive machine learning models (future scope).</li> <li>• Non-child malnutrition (focus only on under-5 children).</li> </ul> </li> </ul> |

| Problem Statement |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description       | Malnutrition among children under five remains a pressing global issue. While raw data is available from multiple sources (compiled on Kaggle), it is fragmented and complex to interpret. Policymakers and stakeholders lack intuitive tools to explore patterns across regions and income groups, which makes it difficult to design effective interventions.                                                                                                                                          |
| Impact            | <p>By solving this problem through an interactive dashboard, decision-makers will gain:</p> <ul style="list-style-type: none"> <li>• Clear visualization of malnutrition patterns over time.</li> <li>• Insights into how malnutrition correlates with economic status.</li> <li>• Ability to prioritize high-risk regions and allocate resources effectively.</li> </ul>                                                                                                                                |
| Proposed Solution |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Approach          | <ul style="list-style-type: none"> <li>• Collect dataset from Kaggle.</li> <li>• Connect datasets to Power BI.</li> <li>• Perform cleaning, transformation, and creation of calculated fields.</li> <li>• Develop multiple visualizations (bar charts, line charts, ribbon charts, etc.) for comparative analysis.</li> <li>• Design an interactive dashboard with filters by country, income level, and malnutrition indicator.</li> <li>• Document findings and record a walkthrough video.</li> </ul> |
| Key Features      | <ul style="list-style-type: none"> <li>• Interactive Power BI dashboard.</li> <li>• Comparative insights between <b>undernutrition</b> and <b>overnutrition</b>.</li> <li>• Multi-decade trend analysis (1983–2019).</li> <li>• Country and income-level filters.</li> <li>• Professional documentation and demonstration video.</li> </ul>                                                                                                                                                              |

## Resource Requirements

| Resource Type       | Description                             | Specification/Allocation                                              |
|---------------------|-----------------------------------------|-----------------------------------------------------------------------|
| <b>Hardware</b>     |                                         |                                                                       |
| Computing Resources | CPU/GPU specifications, number of cores | Standard laptop/PC with Intel i5/i7 CPU                               |
| Memory              | RAM specifications                      | Minimum 8 GB RAM                                                      |
| Storage             | Disk space for data, models, and logs   | 5–10 GB free disk space                                               |
| <b>Software</b>     |                                         |                                                                       |
|                     | Development Tool                        | Power BI Desktop                                                      |
|                     | Additional libraries                    | GitHub for project hosting                                            |
|                     | IDE, version control                    | MS Word / Markdown                                                    |
| <b>Data</b>         |                                         |                                                                       |
|                     | Dataset                                 | Kaggle datasets: malnutrition-estimates.csv, country-wise-average.csv |
|                     | Size                                    | Approx. 11M survey samples                                            |
|                     | Format                                  | CSV                                                                   |